

COS30018 – Intelligent Systems

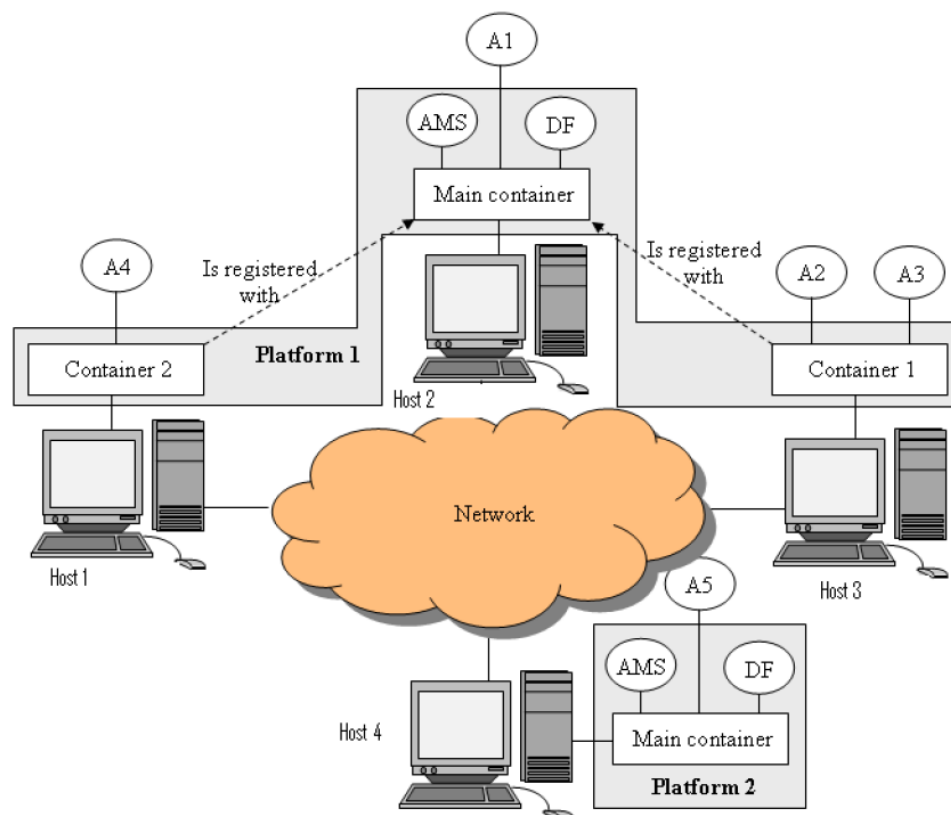
Getting started with JADE

a. Introduction

In this document, we will walk through how to start an example agent with JADE from both the command line and inside the IntelliJ IDE.

JADE architecture contains 4 main components:

- Agent
- Container
- Platform
- Main Container

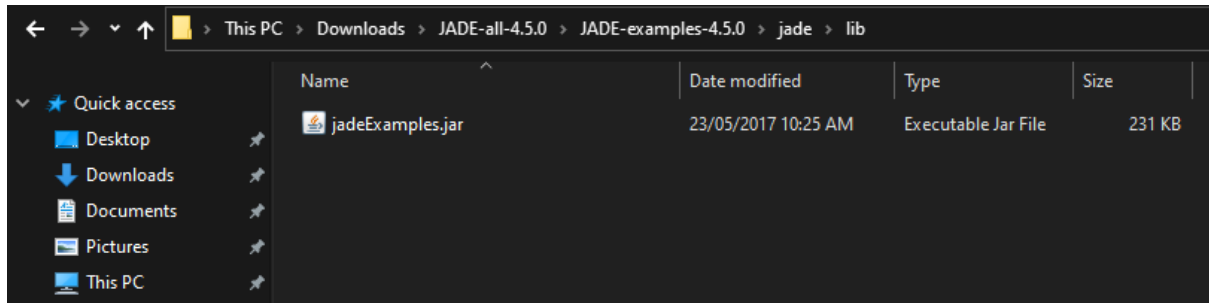


Agents execute tasks and interact by exchanging messages. They live on top of a **Platform** that provides them with basic services such as message delivery. A **Platform** is composed of one or more **Containers**. **Containers** can be executed on different hosts thus achieving a distributed platform. Each **Container** contains zero or more agents. For example, with reference to the figure above, **Container** “Container 1” in host 3 contains **agents** A2 and A3. A special container called **Main Container** exists in each platform. The **Main Container** is itself a container and can therefore contains agents, but differs from other containers as it has two special agents (**AMS** and **DF**).

In this part, we will learn how to create a “HelloWorld” example **agent**. The instructions are given in the file named “JADE example agent” on Canvas. Please follow it to have your first “HelloWorld” **agent** running in the JADE **main container** on your host machine.

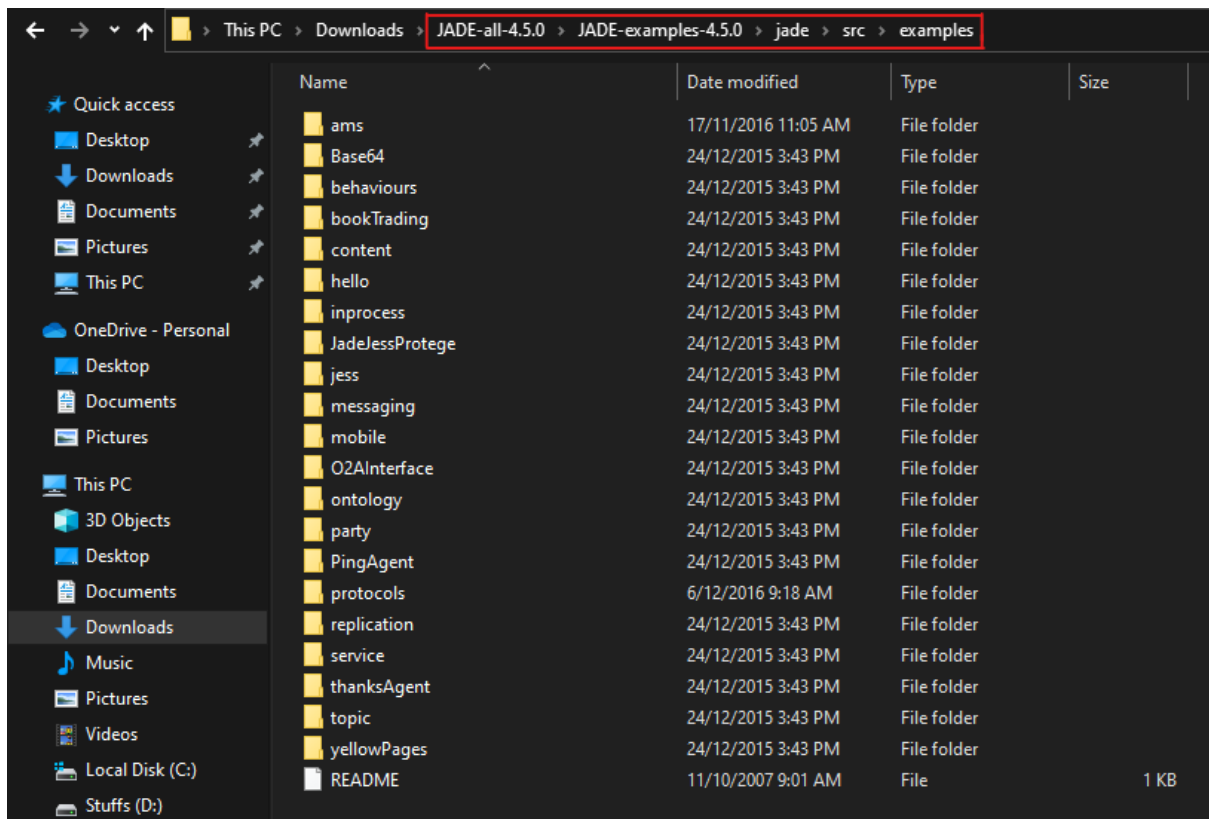
b. Example agent

JADE comes with several example agents that are included in the **JADE-examples-4.5.0.zip** file. Please extract that folder and navigate to the **jade\lib** directory inside. The file **jadeExamples.jar** contains all the examples:



Starting JADE with example agent from command line:

If you navigate to the directory **JADE-all-4.5.0\JADE-examples-4.5.0\jade\src\examples**, you will see there are many example agents that are distributed with the JADE software:



In order to start an example agent, we need to add **jade.jar** and **jadeExample.jar** to the CLASSPATH first. In the screenshot below, we are creating a HelloWorldAgent (examples.hello.HelloWorldAgent) with the name MyAgent. On creation, the agent prints out “Hello world” and then terminates itself:

```

C:\Users\N0 GO BE>cd Downloads\JADE-all-4.5.0
C:\Users\N0 GO BE\Downloads\JADE-all-4.5.0>java -cp JADE-bin-4.5.0\jade\lib\jade.jar;JADE-examples-4.5.0\jade\lib\jadeExamples.jar jade.Boot -gui -agents MyAgent:examples.hello.HelloWorldAgent
JUL 25, 2022 10:35:38 AM jade.core.Runtime beginContainer
INFO: -----
This is JADE 4.5.0 - revision 6825 of 23-05-2017 10:06:04
downloaded in Open Source, under LGPL restrictions,
at http://jade.tilab.com/
-----
JUL 25, 2022 10:35:40 AM jade.imtp.leap.LEAPIMTPManager initialize
INFO: Listening for intra-platform commands on address:
-jicp://192.168.20.98:1099
JUL 25, 2022 10:35:40 AM jade.core.BaseService init
INFO: Service jade.core.management.AgentManagement initialized
JUL 25, 2022 10:35:40 AM jade.core.BaseService init
INFO: Service jade.core.messaging.Messaging initialized
JUL 25, 2022 10:35:40 AM jade.core.BaseService init
INFO: Service jade.core.event.Notification initialized
JUL 25, 2022 10:35:40 AM jade.core.BaseService init
INFO: Service jade.core.resource.ResourceManagement initialized
JUL 25, 2022 10:35:40 AM jade.core.BaseService init
INFO: Service jade.core.mobility.AgentMobility initialized
JUL 25, 2022 10:35:40 AM jade.core.BaseService init
INFO: Service jade.core.event.Notification initialized
JUL 25, 2022 10:35:40 AM jade.mtp.http.HTTPServer <init>
INFO: HTTP-MTP Using XML parser com.sun.org.apache.xerces.internal.jaxp.SAXParserImpl$JAXPSAXParser
JUL 25, 2022 10:35:40 AM jade.core.messaging.MessagingService boot
INFO: MTP addresses:
http://192.168.56.1:7778/acc
Hello World! My name is MyAgent
JUL 25, 2022 10:35:40 AM jade.core.AgentContainerImpl joinPlatform
INFO: -----
Agent container Main-Container@192.168.20.98 is ready.
-----

```

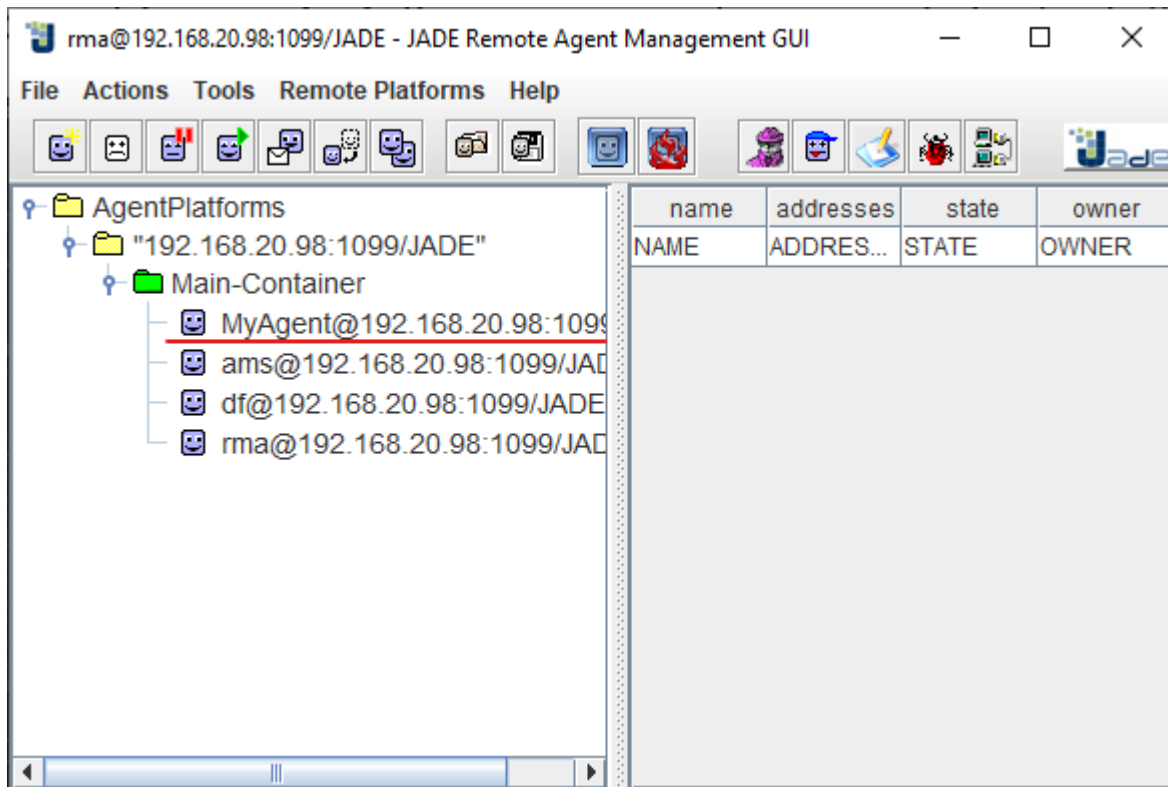
All example agent classes belong to subpackages of the **‘examples’** package, refer to the various example agents with a fully qualified package name; for example, class **ComplexBehaviourAgent** in **‘behaviours’** subdirectory of **‘examples’** directory must be started via the following command line:

```

C:\Users\N0 GO BE>cd Downloads\JADE-all-4.5.0
C:\Users\N0 GO BE\Downloads\JADE-all-4.5.0>java -cp JADE-bin-4.5.0\jade\lib\jade.jar;JADE-examples-4.5.0\jade\lib\jadeExamples.jar jade.Boot -gui -agents MyAgent:examples.behaviours.ComplexBehaviourAgent
JUL 25, 2022 10:47:06 AM jade.core.Runtime beginContainer
INFO: -----
This is JADE 4.5.0 - revision 6825 of 23-05-2017 10:06:04
downloaded in Open Source, under LGPL restrictions,
at http://jade.tilab.com/
-----
JUL 25, 2022 10:47:08 AM jade.imtp.leap.LEAPIMTPManager initialize
INFO: Listening for intra-platform commands on address:
-jicp://192.168.20.98:1099
JUL 25, 2022 10:47:08 AM jade.core.BaseService init
INFO: Service jade.core.management.AgentManagement initialized
JUL 25, 2022 10:47:08 AM jade.core.BaseService init
INFO: Service jade.core.messaging.Messaging initialized
JUL 25, 2022 10:47:08 AM jade.core.BaseService init
INFO: Service jade.core.event.Notification initialized
JUL 25, 2022 10:47:08 AM jade.core.BaseService init
INFO: Service jade.core.resource.ResourceManagement initialized
JUL 25, 2022 10:47:08 AM jade.core.BaseService init
INFO: Service jade.core.mobility.AgentMobility initialized
JUL 25, 2022 10:47:08 AM jade.core.BaseService init
INFO: Service jade.core.event.Notification initialized
JUL 25, 2022 10:47:08 AM jade.mtp.http.HTTPServer <init>
INFO: HTTP-MTP Using XML parser com.sun.org.apache.xerces.internal.jaxp.SAXParserImpl$JAXPSAXParser
JUL 25, 2022 10:47:08 AM jade.core.messaging.MessagingService boot
INFO: MTP addresses:
http://192.168.56.1:7778/acc
Agent MyAgent@192.168.20.98:1099/JADE: Step 1.1
Agent MyAgent@192.168.20.98:1099/JADE: Step 2.1.1
JUL 25, 2022 10:47:08 AM jade.core.AgentContainerImpl joinPlatform
INFO: -----
Agent container Main-Container@192.168.20.98 is ready.
-----
Agent MyAgent@192.168.20.98:1099/JADE: Step 1.2
Agent MyAgent@192.168.20.98:1099/JADE: Step 2.1.2
Agent MyAgent@192.168.20.98:1099/JADE: Step 1.3
Agent MyAgent@192.168.20.98:1099/JADE: Step 2.1.3
Agent MyAgent@192.168.20.98:1099/JADE: Step 2.2.1
Agent MyAgent@192.168.20.98:1099/JADE: Step 2.2.2
Agent MyAgent@192.168.20.98:1099/JADE: Step 2.2.3
Agent MyAgent@192.168.20.98:1099/JADE: Step 2.3
Agent MyAgent@192.168.20.98:1099/JADE: Step 2.4
Agent MyAgent@192.168.20.98:1099/JADE: Step 2.5

```

You can see the new agent is living in the main container by seeing the GUI:

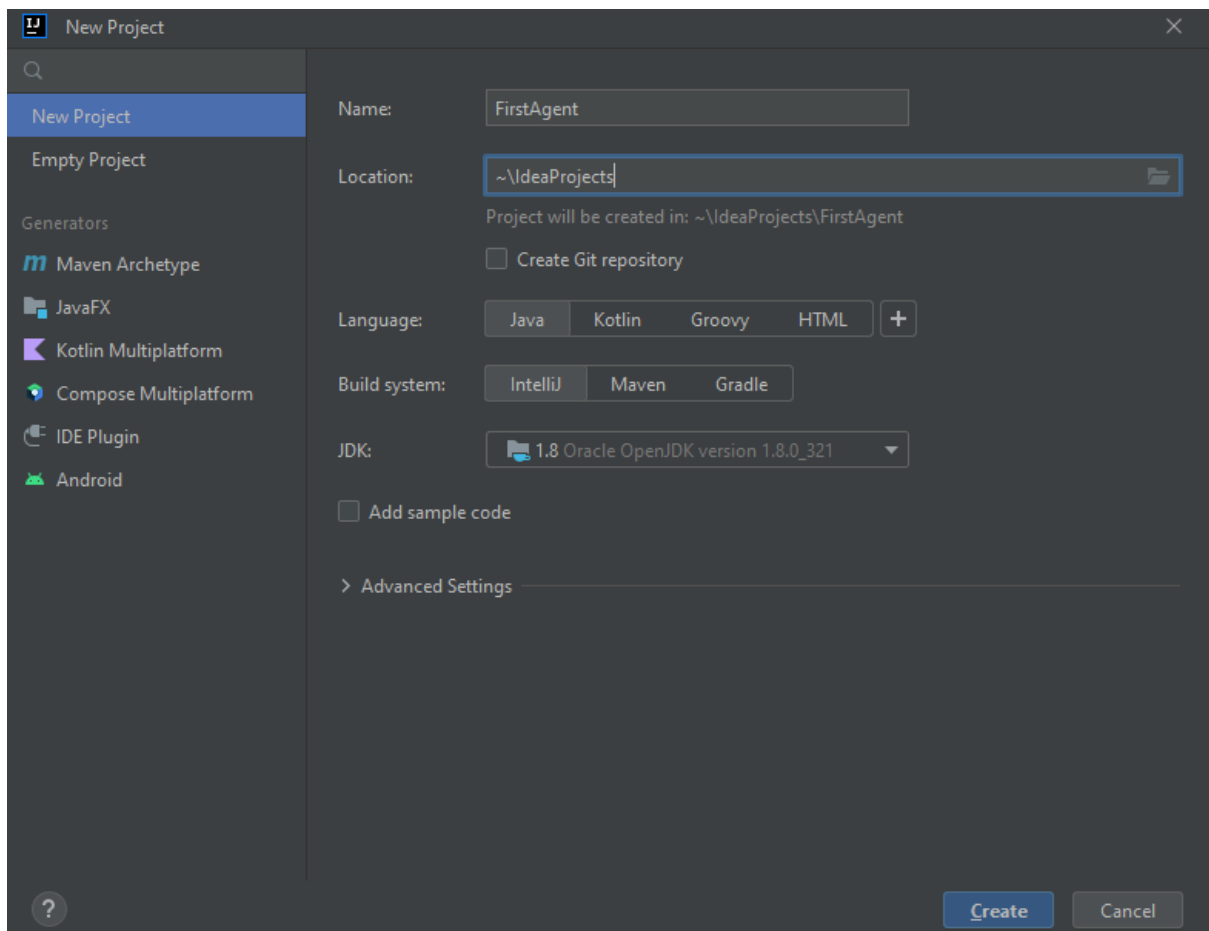


NOTE: Please terminate the JADE platform before restarting it since the platform can have only one main container

Runnig JADE from inside IntelliJ:

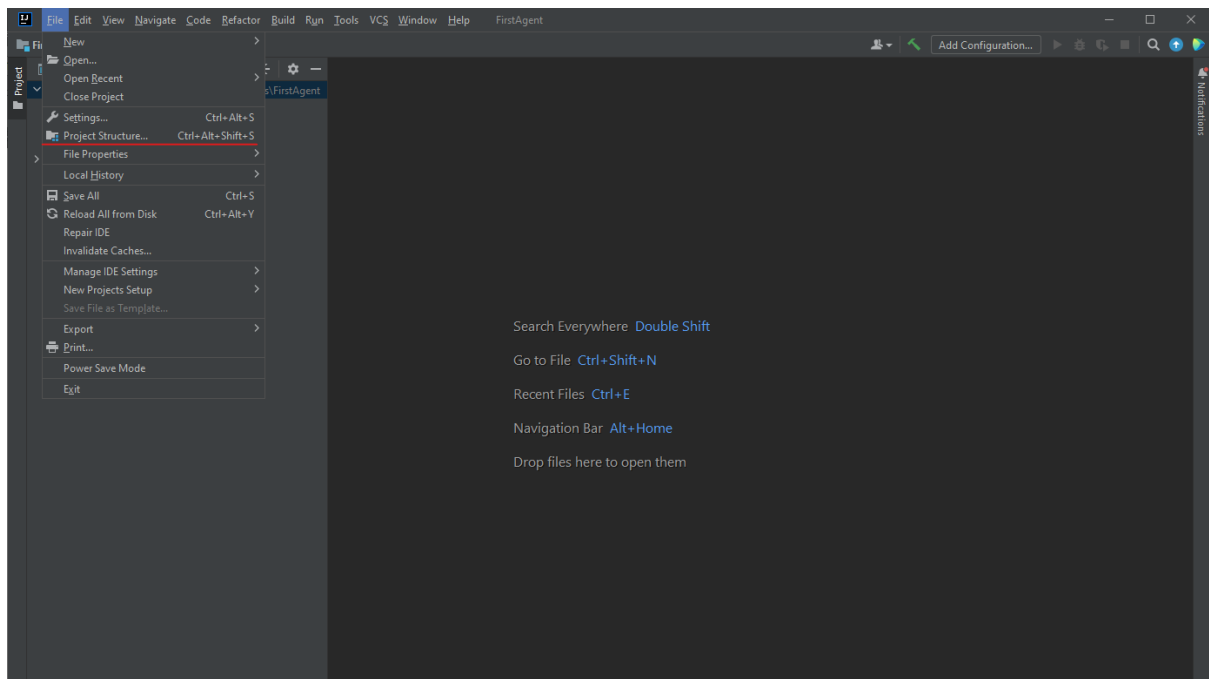
In the previous examples, we ran JADE from the command line. However, we can do the same from inside an IDE such as IntelliJ. The steps below show how to create a Java Project in IntelliJ and run JADE from within the IDE.

Step 1: Create a new project. Start IntelliJ and create a new project, give it a name and the location where you want to save it:

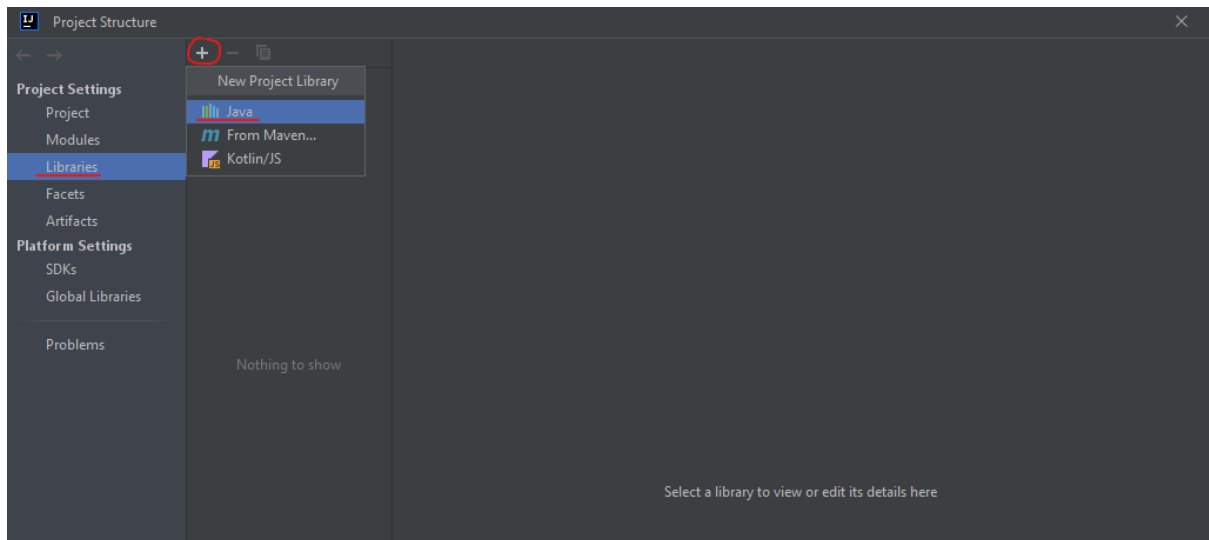


Step 2: Add JADE libraries to the project

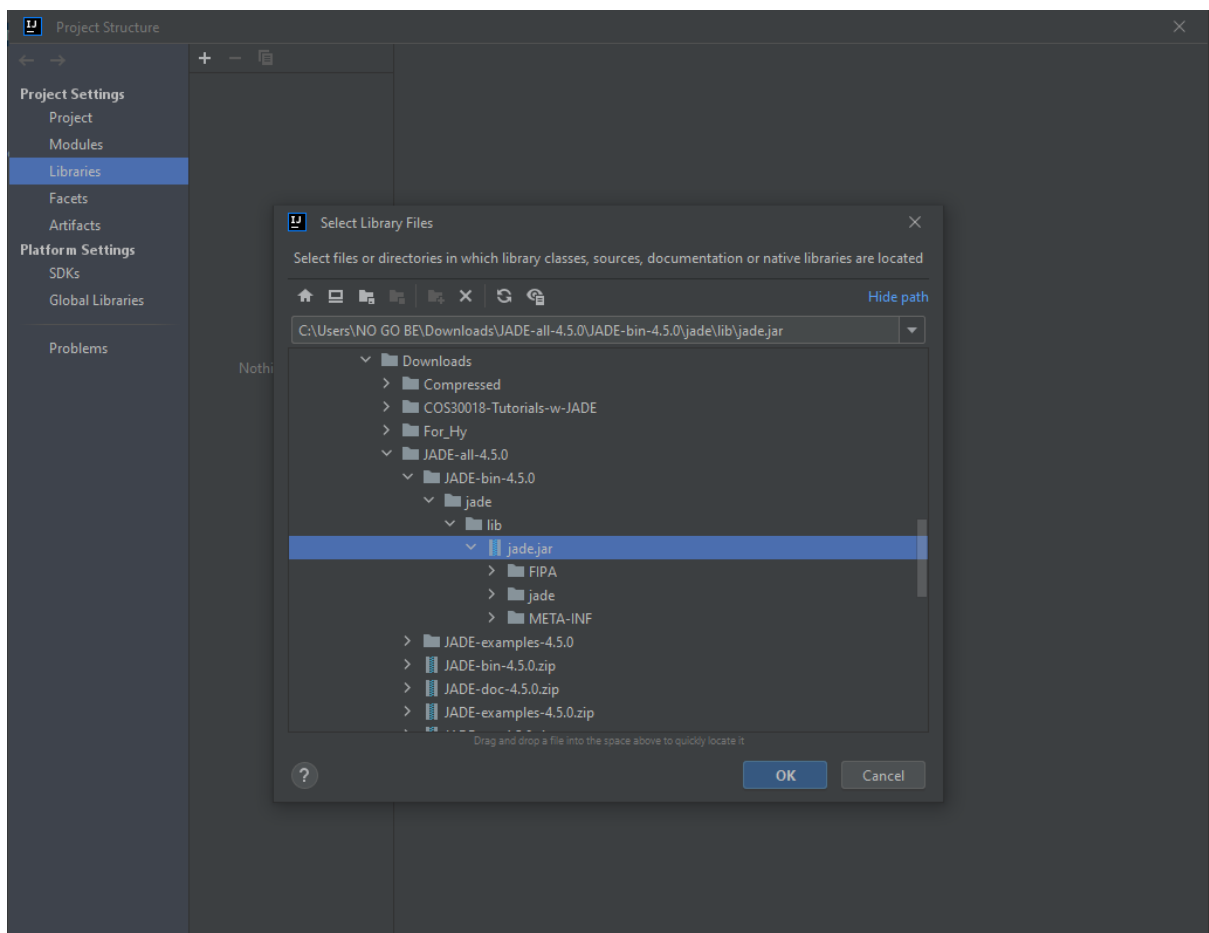
Go to File -> Project Structure and choose “Libraries” under the “Project Settings” to create a new folder for your JADE libraries:

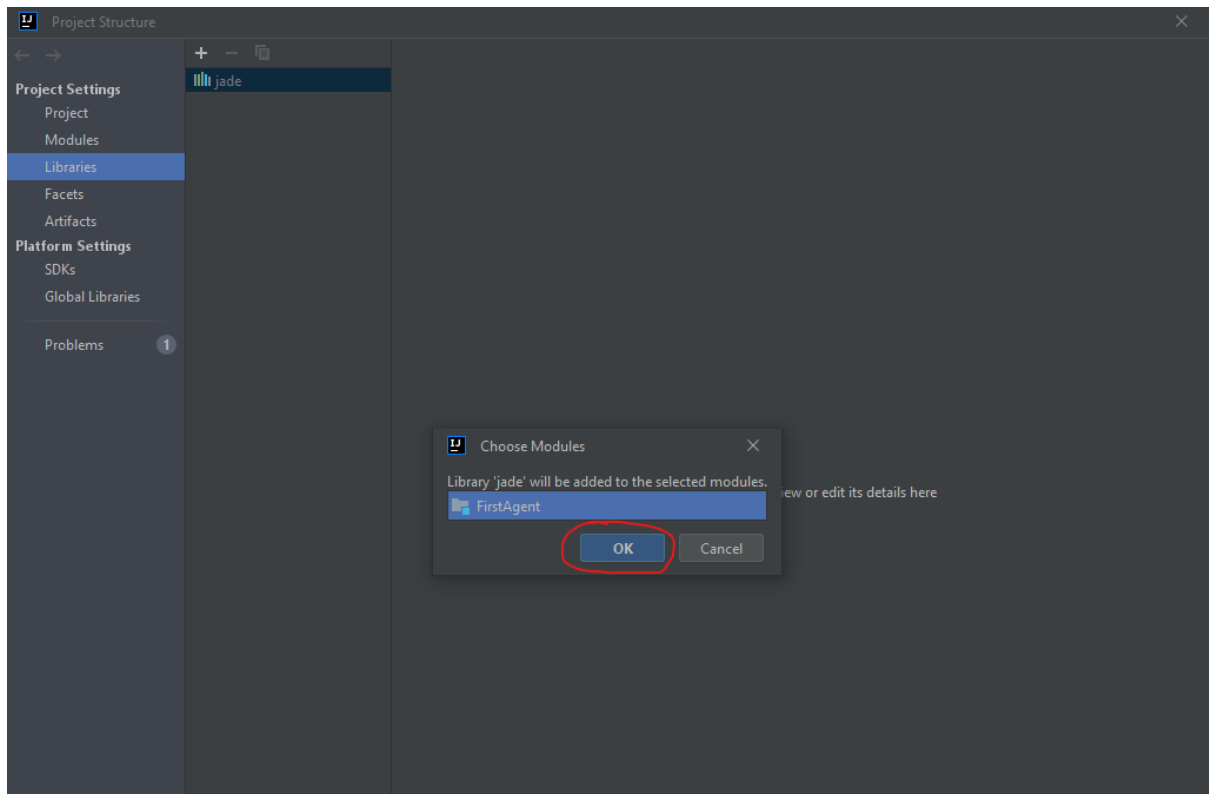


Click on the “+” button and choose “Java” to add a Java library:

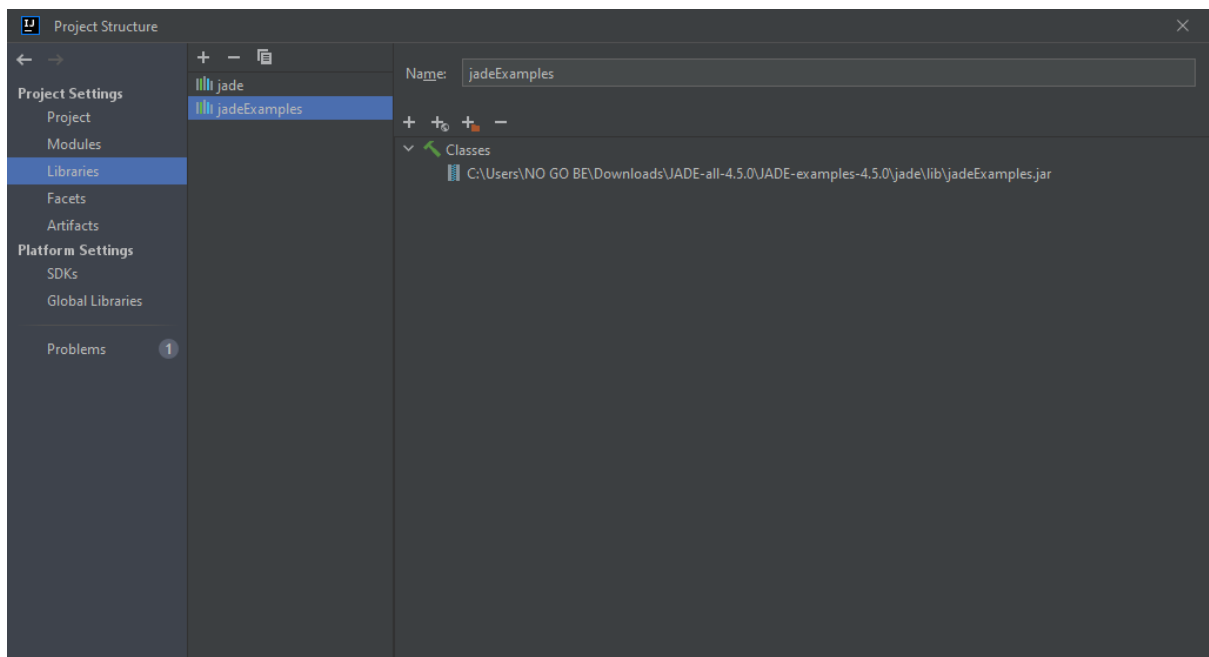


Then navigate to the location where you keep your “**jade.jar**” file and add it:



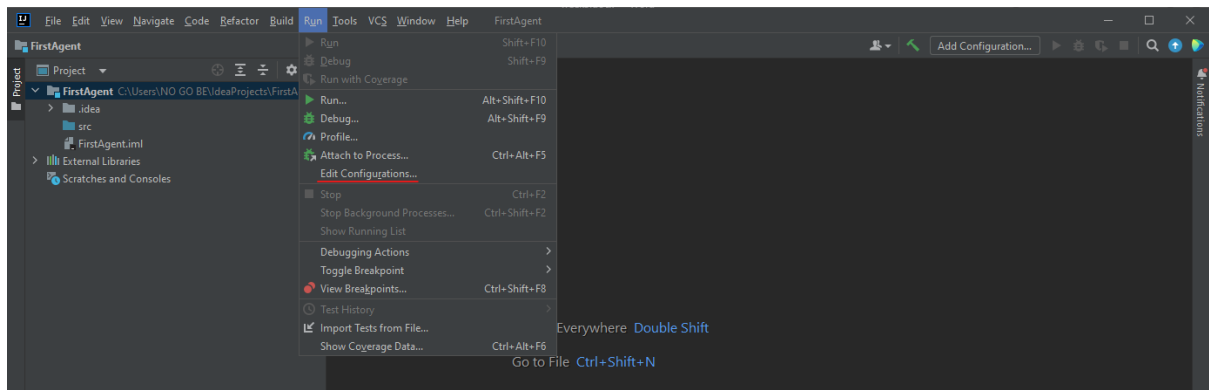


Then, do it similarly to add the file “**jadeExamples.jar**”:

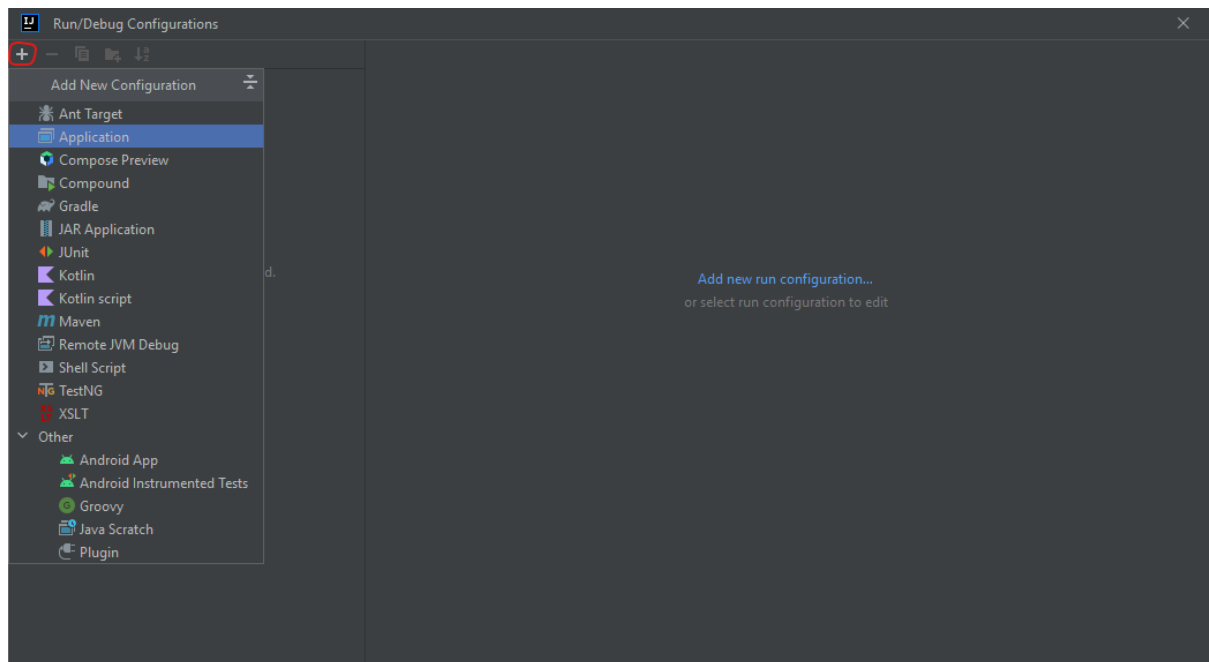


Step 3: Configure the project to run JADE from within the IDE

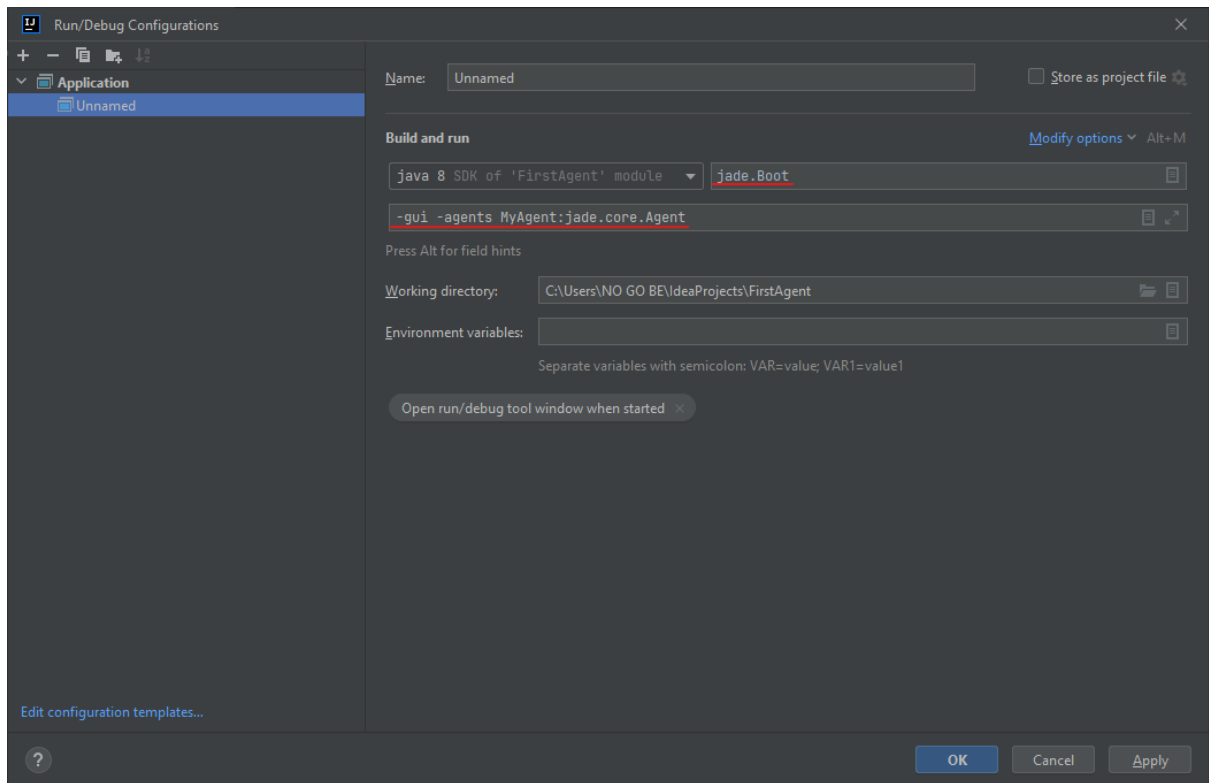
Go to “run” and choose “Edit Configurations”:



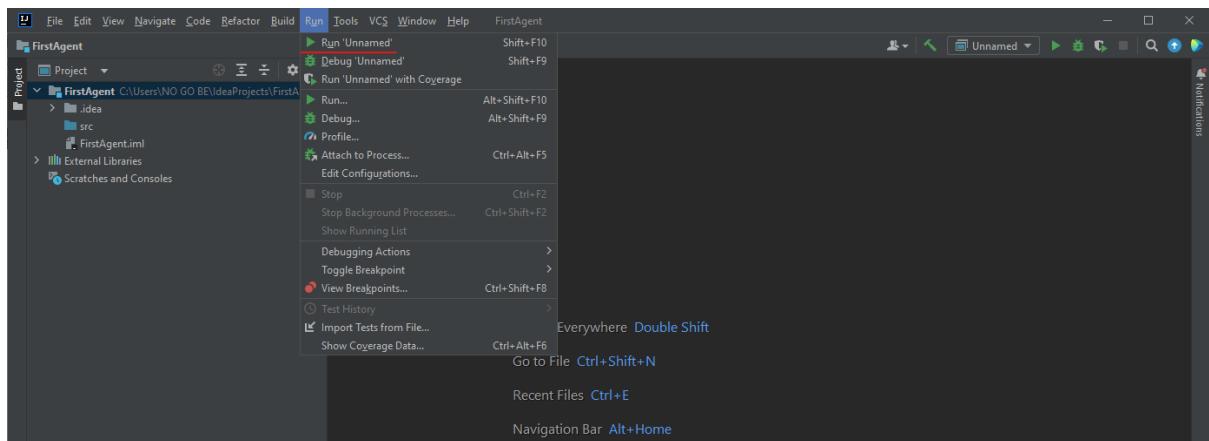
Click on “Add New Configuration” and choose “Application”:



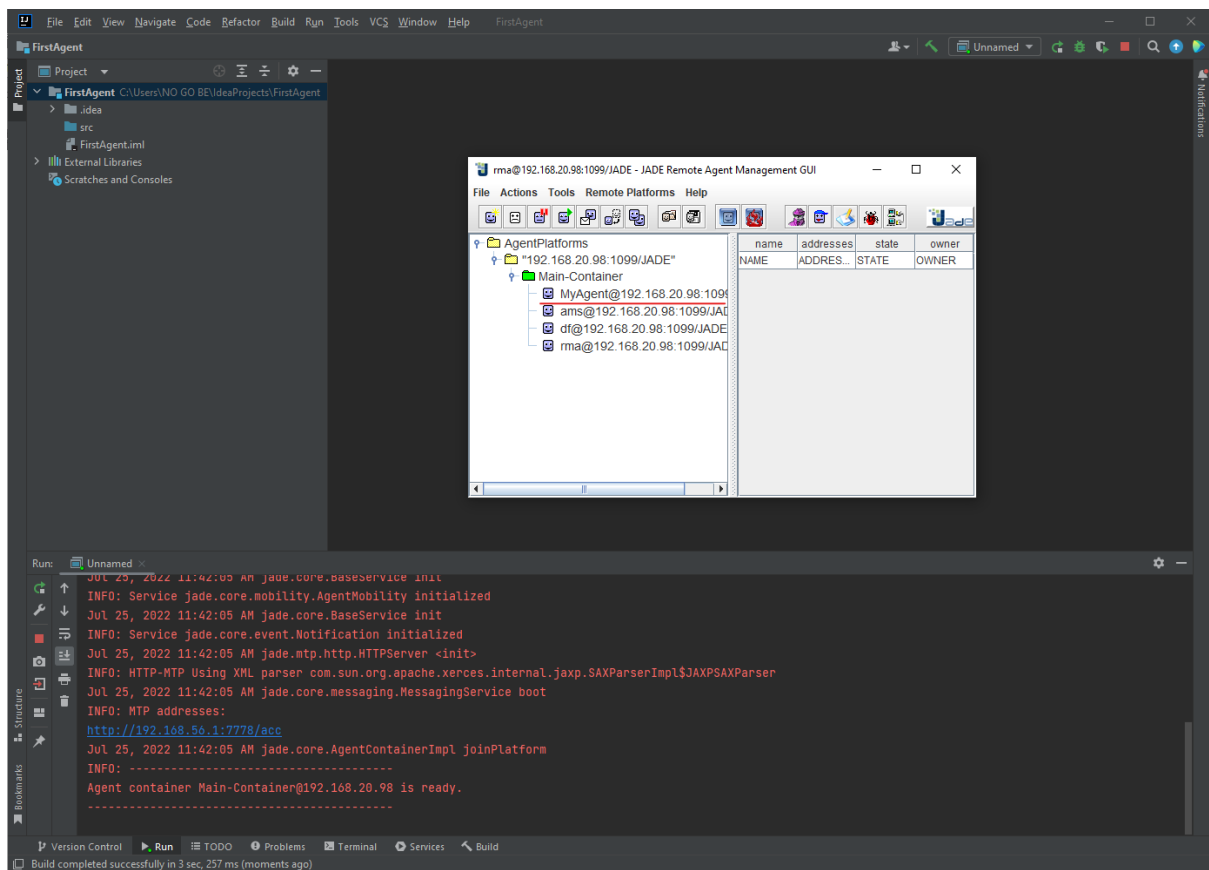
Then we configure the main class as **jade.Boot** and specify any arguments that the Boot class takes (e.g. **-gui** to start the Graphical User interface). If we want to start up MyAgent, we can specify that as well using the **-agents** flag. The naming convention is **<AgentName>:<Agent Class>** i.e. **MyAgent:jade.core.Agent**:



Apply the configuration to save it. You can now run it to start up JADE:



If you have typed in everything correctly, you should be able to see the JADE Remote Agent Management GUI and the MyAgent within the Main Container as below:



Congratulations! You have successfully configured your IntelliJ IDE to launch the JADE platform within a self-contained JAVA project. Now you can start writing your own “intelligent agents”.